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Q. Prove that the set of rational numbers $\mathbb{Q}$. equipped with the two binary operations of addition & multiplication, forms a field.

→ we take the rational numbers $\mathbb{Q}$ to be the set of equivalance classes of ordered pairs (a,b) with $a,b \in \mathbb{Z}$ and $b \neq 0$, where $(a,b) \sim (a',b')$ iff $ab' = a'b$, we identify the class of (a,b) with the usall fraction $\frac{a}{b}$. Define addition & multiplication in the usal way:

$$\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}, \quad \frac{ac}{bd} = \frac{a}{b} \cdot \frac{c}{d}.$$

for $b \neq 0$, $d \neq 0$, below we show those operation make $\mathbb{Q}$ a field.

1. The operations are well defined.

we must check that if $\frac{a}{b} = \frac{a'}{b'}$ and $\frac{c}{d} = \frac{c'}{d'}$

then $\frac{ad+bc}{bd} = \frac{a'd'+b'c'}{b'd'}$ & $\frac{ac}{bd} = \frac{a'c'}{b'd'}$

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$(ad+bc)b'd' = (ab')(dd') + (bc)(b'd')$

$= (a'b)(dd') + (bc)(bd)$:

So similarly expand the right hand numerator times $bd\,b'd'$. Rearranging and using $ab' = a'b$, $ad' = c'd$. Shows both cross-product are equal therefore the sums represent the same equivalence

2. $(\mathbb{Q}, +)$ is an abelian group.

Take any $\frac{a}{b}, \frac{c}{d}, \frac{e}{f} \in \mathbb{Q}$.

Closure : $\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$ is a rational number.

Associativity : $\left(\frac{a}{b} + \frac{c}{d}\right) + \frac{e}{f} = \frac{ad+bc}{db} + \frac{e}{f}$

$= \frac{f(ad+bc) + e(bd)}{(bd)f}$ & a

Similar expansion for $\frac{a}{b} + \left(\frac{c}{d} + \frac{e}{f}\right)$:

Identify :- $0 = \frac{0}{1}$ satisfy.

Commutativity :- $\frac{a}{b} + \frac{c}{d} = \frac{cd+bc}{bd}$, $\frac{bc+ad}{bd}$,

$\frac{c}{d} + \frac{a}{b}$.

Thats $(Q, +)$ is an abelian group.

3. multiplication on $Q/\{0\}$ is an abelian group.

Closure : Product $\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$ is rational since $bd \neq c$.

Associativity & commutativity :-

$$\left(\frac{a}{b} \cdot \frac{c}{d}\right) \frac{e}{f} = \frac{ac}{bd} \cdot \frac{e}{f} = \frac{(ac)e}{bdf} = \frac{ace}{dfb}$$

$$= \frac{a}{b} \cdot \frac{ce}{df} = \frac{a}{b}\left(\frac{c}{d} \cdot \frac{e}{f}\right)$$

So, Q is a commutative ring with unity 1.